Project Title: Early Detection of High-Risk Diagnoses Leading to Inpatient Readmissions

1. Business Problem

Hospital readmissions are a pressing issue for healthcare systems globally. They not only lead to increased healthcare costs but also negatively impact patient outcomes and hospital performance metrics. Readmission rates are closely monitored by regulatory bodies, and hospitals with high rates can face significant financial penalties. The goal of this project is to leverage machine learning to build a predictive model that can identify patients at high risk of readmission early, particularly those diagnosed with diabetes. By detecting risk factors during or prior to discharge, healthcare providers can implement targeted interventions that reduce readmission rates and improve patient care.

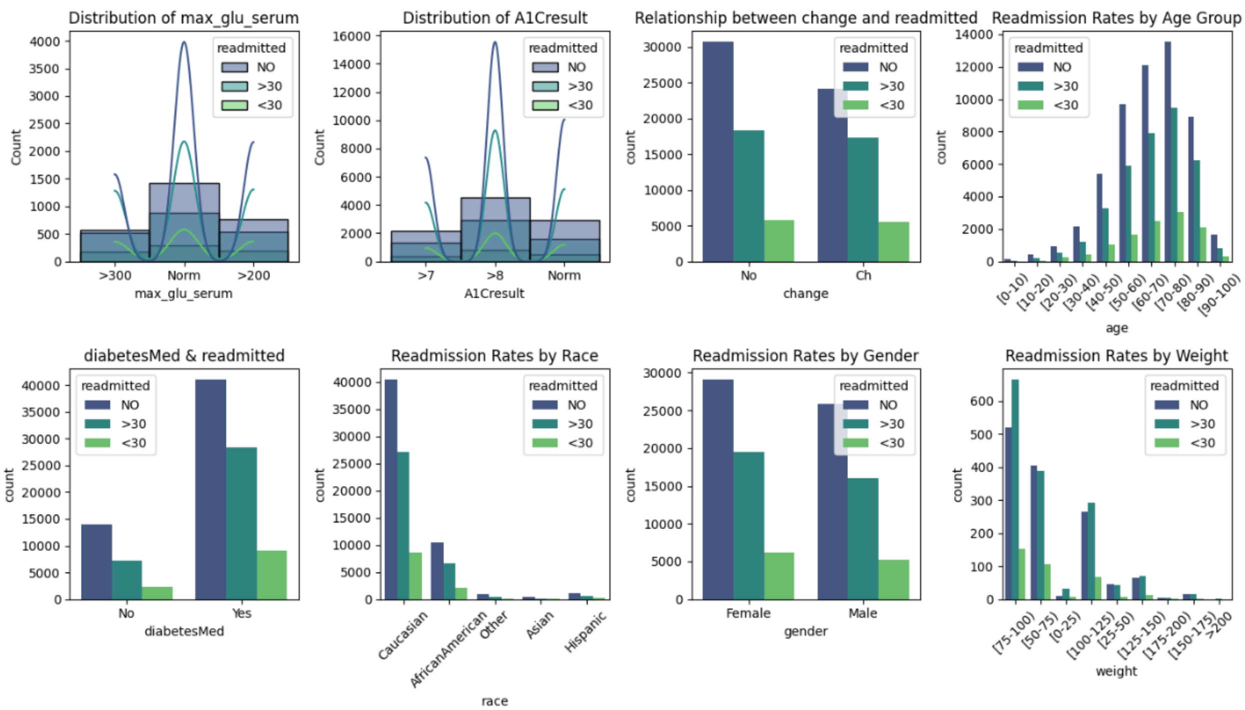
2. Background/History

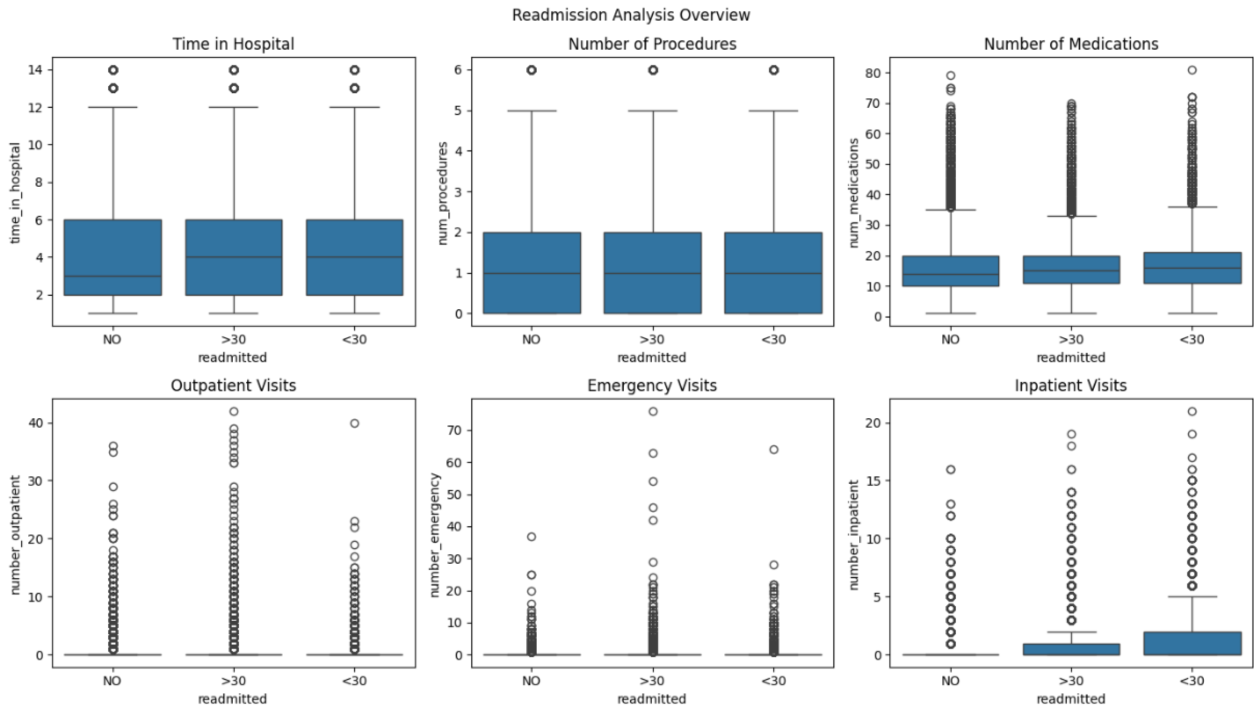
Unplanned hospital readmissions within 30 days have become a focal point for healthcare quality improvement. The Centers for Medicare and Medicaid Services (CMS) penalizes hospitals with high readmission rates under the Hospital Readmissions Reduction Program (HRRP). Chronic conditions such as diabetes are among the leading causes of such readmissions. The increased adoption of electronic health records (EHR) presents an opportunity to apply data science methods to historical patient data to uncover patterns associated with readmissions. Previous efforts have often focused on rule-based systems or logistic regression models, but the complexity and non-linearity of patient data necessitate more advanced approaches such as ensemble machine learning techniques.

3. Data Explanation

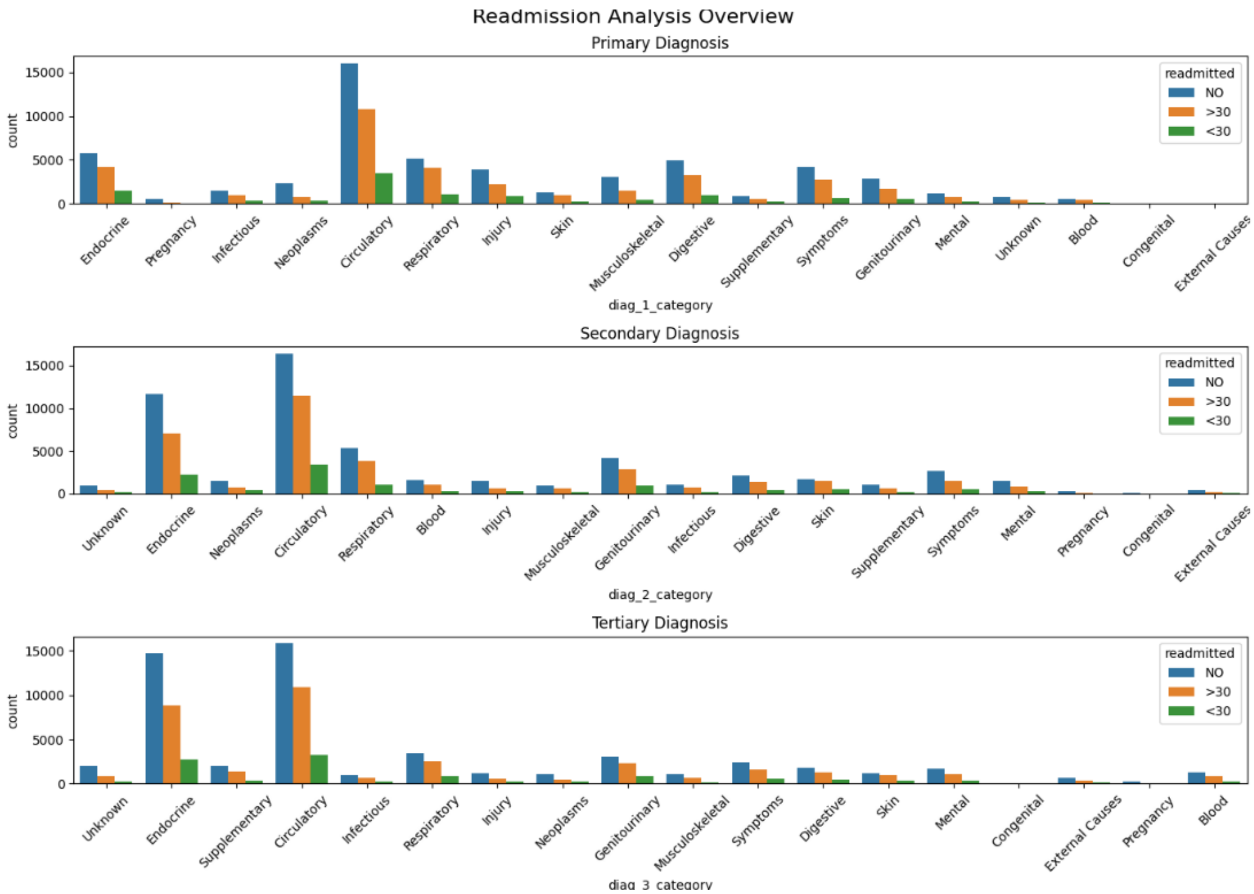
The dataset used in this project is the Diabetes 130-US hospitals dataset, which covers patient records from 1999 to 2008 and is available from the UCI Machine Learning Repository. It includes over 100,000 records and contains features such as demographics, diagnoses, lab results, medications, and hospital utilization.

To prepare the data, I cleaned and preprocessed it, removing rows with missing values in key fields and converting categorical data into numerical values. I also filtered the dataset to keep only one encounter per patient to avoid redundancy. Since the data was imbalanced in terms of readmission outcomes, I applied balancing techniques to ensure fair model training.





Key variables used in the analysis include the target variable readmitted, which indicates whether a patient was readmitted within 30 days. Some of the predictor variables include lab test results such as max_glu_serum and A1Cresult, various diabetes-related medications, prior hospital utilization metrics like number_inpatient and number_emergency, and demographic attributes including age, gender, and race.



4. Methods

I tested four machine learning algorithms: Logistic Regression, Random Forest, Neural Network, and XGBoost. My evaluation focused on accuracy, precision, recall, F1-score, and ROC-AUC to ensure a comprehensive assessment.

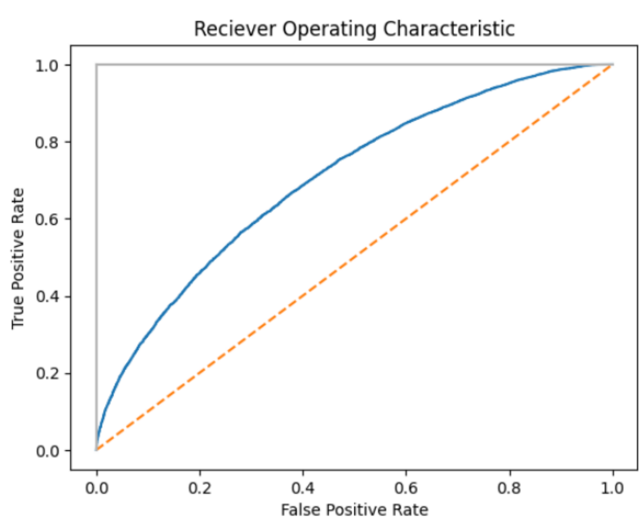
XGBoost turned out to be the best-performing model, and I was particularly impressed with its ability to capture complex patterns in tabular healthcare data. To make the model interpretable, I used SHAP (Shapley) values. These helped me understand which features were influencing predictions the most and provided transparency that clinicians can trust.

5. Analysis

After evaluating all four models, I observed that the XGBoost model outperformed both Logistic Regression and Random Forest in predicting inpatient readmissions. The XGBoost model achieved a higher ROC-AUC score (0.7042) and a better balance between precision (0.63) and recall (0.56), resulting in a stronger F1-score of 0.60 for class 1 (readmitted). These metrics suggest that while the model performs reasonably well, it still misses a significant number of readmitted cases, which is a known challenge in imbalanced medical datasets. The ROC curve shown below confirmed that the model performed better than random, with the curve staying above the diagonal line.

```
Confusion Matrix: XGBoost
 [[7914 3058]
 [4106 5275]]
Accuracy Score: 0.65
ROC-AUC Score: 0.7041918934429056
              precision    recall  f1-score   support

           0       0.66      0.72      0.69     10972
           1       0.63      0.56      0.60      9381

    accuracy                           0.65     20353
   macro avg       0.65      0.64      0.64     20353
weighted avg       0.65      0.65      0.65     20353
```

The XGBoost model achieved an overall accuracy of 65% and a ROC-AUC score of 0.70, indicating good discrimination between patients who were readmitted and those who were not. The confusion matrix revealed that the model correctly predicted 7,914 non-readmitted cases and 5,226 readmitted cases, with 3,058 false positives and 4,106 false negatives.

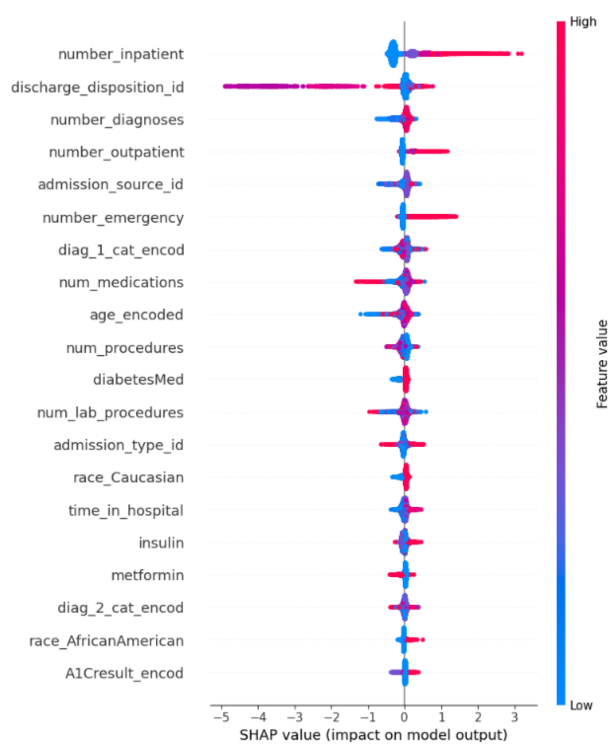


In contrast, both the Logistic Regression and Random Forest models performed similarly, with an accuracy of 0.62 and an F1-score of 0.53 for the readmission class. Their ROC-AUC scores were 0.6475 and 0.6847, respectively, lower than XGBoost's.

```
Confusion Matrix: Logistic Regression
 [[8298 2674]
 [4993 4388]]
Accuracy Score: 0.62
ROC-AUC Score: 0.6475430593784421
              precision    recall  f1-score   support

           0       0.62      0.76      0.68     10972
           1       0.62      0.47      0.53      9381

    accuracy                           0.62     20353
   macro avg       0.62      0.61      0.61     20353
weighted avg       0.62      0.62      0.61     20353
```

```
Confusion Matrix: Random Forest
 [[8298 2674]
 [4993 4388]]
Accuracy Score: 0.62
ROC-AUC Score: 0.6847182659095263
              precision    recall  f1-score   support

           0       0.62      0.76      0.68     10972
           1       0.62      0.47      0.53      9381

    accuracy                           0.62     20353
   macro avg       0.62      0.61      0.61     20353
weighted avg       0.62      0.62      0.61     20353
```

Using SHAP values, I uncovered that the most influential features were the number of prior inpatient visits and the discharge disposition. Patients with frequent recent inpatient visits or specific discharge dispositions were at a significantly higher risk of being readmitted. Features like number of outpatient, emergency, and diagnosis encounters also contributed positively to risk prediction. Meanwhile, demographic and medication-related features such as race, insulin, and metformin usage had relatively lower impact, suggesting limited bias or reduced relevance in this context.



Further statistical analysis also showed that max_glu_serum levels differed significantly between the readmitted and non-readmitted groups (p-value < 1e-8), highlighting its importance as a predictor. Medication usage patterns also revealed interesting associations. The use of insulin, metformin, glipizide, and rosiglitazone was significantly correlated with readmission rates, while others like nateglinide and acetohexamide were not

6. Conclusion

The project successfully demonstrated the potential of machine learning to predict inpatient readmissions using historical EHR data. The XGBoost model, combined with SHAP explanations, provides a robust and interpretable framework for identifying high-risk patients. This approach can empower healthcare providers to take proactive measures and improve both operational efficiency and patient outcomes.

The predictive modeling approach revealed valuable insights into readmission risk factors. While none of the models achieved high precision, the XGBoost model’s superior performance demonstrates its potential to assist in clinical decision-making. I was able to highlight key predictors like prior inpatient visits and discharge disposition that hospitals can use to proactively identify high-risk patients. These findings not only support earlier intervention but also guide the allocation of care management resources more effectively.

Despite moderate accuracy, the model’s consistent performance indicates reliability. When used as part of a larger risk management strategy, it can help reduce readmissions and improve patient outcomes.

7. Assumptions

The model assumes that historical data patterns are still relevant and applicable to current patient populations. It also presumes the accuracy and consistency of recorded data such as lab values and medication usage. Furthermore, the analysis relies on the assumption that patients received similar quality of care across different hospitals included in the dataset.

8. Limitations

While the model performs reasonably well, its 65% accuracy may not be sufficient for fully automated clinical deployment without human oversight. The dataset's class imbalance, with fewer readmitted patients, may have affected model sensitivity.

9. Challenges

Balancing the trade-off between model complexity and interpretability was another hurdle. Ensuring that the model's predictions remain valid across different hospital settings and patient demographics also posed a challenge.

10. Future Uses/Additional Applications

The developed model can be integrated into hospital EHR systems to flag high-risk patients prior to discharge. This early warning system can prompt timely interventions such as follow-up appointments, home care referrals, or medication adjustments. Furthermore, incorporating real-time data feeds could enable dynamic risk scoring and more responsive care strategies.

11. Recommendations

It is recommended that hospitals conduct real-world testing of the model in a controlled pilot environment. Additionally, combining model predictions with case manager follow-ups can help mitigate false negatives and ensure patients receive appropriate care post-discharge.

12. Implementation Plan

The implementation will proceed in multiple phases. Phase 1 involves deploying the model on historical patient data from a single hospital to validate its performance. In Phase 2, the model will be integrated with live EHR systems to begin flagging high-risk patients in real-time Finally, Phase 3 will involve scaling the system to additional departments and hospitals within the network.

13. Ethical Assessment

Ethical considerations include ensuring the transparency and fairness of the model. The use of SHAP values enhances transparency by providing clear explanations for each prediction. Care must be taken to avoid algorithmic bias, especially against underrepresented groups. Importantly, the model should support, not replace, clinical decision-making to maintain trust and accountability in patient care.

References

- UCI Machine Learning Repository: Diabetes 130-US hospitals dataset documentation
- Peer-reviewed journals on hospital readmission prediction and chronic disease modeling
- CDC and WHO diabetes treatment guidelines
- SHAP and LIME documentation for model explainability in healthcare